VICTORY OROBOSA

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EDUCATION

University of Central Arkansas – Conway, AR *Expected May 2027*
B.S. Computer Science, Specialization in Software Engineering **GPA: 3.78 / 4.0**
Coursework: Data Structures & Algorithms, Operating Systems, Databases, Artificial Intelligence / Machine Learning, Data Clustering, Principles of Programming Languages, Discrete Mathematics
Honors: Presidential Scholar (4x) • Global Citizen Scholar (2x) • Foundation Scholar (2x) • Dean's List

TECHNICAL SKILLS

Languages & Frontend: Python, C++, TypeScript/JavaScript, Java, SQL, Kotlin, HTML/CSS; React, Vite, TanStack Query, Recharts
Backend & Data: FastAPI, PostgreSQL, SQLAlchemy, Alembic, Celery, Redis, REST APIs, Node.js, Pandas, NumPy
Systems: C++20, CMake, data structures & algorithms, performance benchmarking, memory profiling
Infrastructure & Testing: Docker, GitHub Actions (CI/CD), Git, pytest, Vitest, doctest, ASan / UBSan, Prometheus, Linux, Netlify / Render
AI / ML: PyTorch, Gymnasium, reinforcement learning; Anthropic Claude API, tool use / agents, structured outputs, LLM testing

PROJECTS

AI Supply Chain Intelligence Platform | Python, FastAPI, PostgreSQL, Celery, React/TS, Claude API
Turns any shipment CSV into computed KPIs, a 0–100 risk score, and plain-English explanations. *Live Demo* | *Source Code*

- Built and deployed a multi-tenant analytics platform: 26 REST endpoints on FastAPI with JWT / bcrypt auth, PostgreSQL with SQLAlchemy and Alembic migrations, a Celery + Redis queue, and a React 18 + TypeScript dashboard.
- Built a custom tool-using agent (no framework): the model picks from 9 tools over the analytics layer across up to 6 steps, with tenancy enforced in the tools rather than the prompt and every call shown in the UI for audit.
- Kept every figure deterministic by computing metrics in Python and using the LLM only to explain them; metrics persist before the model call, so the dashboard survives an LLM outage.
- Cut a 300K-row ingest from 84s to 14s by caching a date-format search that was raising ~3M exceptions, and held memory flat on 100 MB uploads by streaming rows in 5,000-record batches.
- Mapped arbitrary vendor exports onto a ~50-field canonical schema using ~400 column aliases plus a fuzzy-match fallback.
- Automated CI in GitHub Actions (379 tests, Postgres migration round-trip, Docker smoke test); fixed a Redis bug CI exposed, where connections outlived the event loop that owned them and crashed Celery tasks.
- Instrumented for Prometheus (HTTP latency, rows ingested, analysis outcomes, LLM tokens), labelling by route template to bound cardinality; readiness returns 503 when Postgres is down, Redis degrades gracefully.

SearchFlix – Movie Search Engine & Index Benchmark | C++20, React/TS, GitHub Actions Source Code

- Benchmarked 7 index structures (BST, AVL, hash table, sorted array vs. 2 STL baselines) over 12.4M IMDb titles: hashing found titles in 91 ns vs ~2 µs for trees but can't autocomplete; a sorted array autocompleted fastest (1.9 µs) at 27 ms per insert.
- Made the benchmark reproducible (batched nanosecond timing, seeded tiers, exact heap accounting via a replaced *operator new*); traced 2× run-to-run drift to hybrid-CPU scheduling, and core pinning raised agreement from 26 to 123 of 132.
- Packed 12.4M titles into 625 MB (from ~2.5 GB) with a 32-byte record; joined IMDb episode data for 11.7M searchable keys, and shipped a C++ HTTP server, React UI, 104 tests, and CI on MSVC, GCC and Clang with sanitizers.

Double DQN – Lunar Lander | Python, PyTorch, Gymnasium, pytest Source Code

- Trained a DQN agent (experience replay, soft target updates, Huber loss) on LunarLander-v3; in a matched-seed ablation vanilla DQN solved 3/3 seeds (272.4 ± 1.4) while Double DQN diverged on 2 of 3 (−453.8 vs −177.6 random).
- Rebuilt a 3-person course notebook into a reproducible repo: checkpoints, seeded 100-episode evaluation, per-run JSON results, 42 tests on the Bellman target and target updates, and CI running a real training smoke test.

EXPERIENCE

Resident Assistant – University of Central Arkansas, Conway, AR *Aug 2024 – Present*

- Support 250+ residents, mediating roommate and policy conflicts and handling after-hours incident response.
- Plan and run 10+ educational and social programs a year, coordinating budget, logistics, and promotion.

Supplemental Instruction Leader / Tutor – University of Central Arkansas, Conway, AR *Aug 2023 – Aug 2024*

- Led 50+ study sessions for 20+ students in CS I/II, Data Structures, Calculus, Physics and Biology, building worked examples and targeted exam reviews; program-reported pass rates rose 25%.

LEADERSHIP

Project Lead, SafeMed Capstone – University of Central Arkansas *Aug 2026 – Present*

- Lead a 3-person team building a medication-safety and caregiver-alert system; own architecture, scope and safety-check logic.

President, African Student Association – University of Central Arkansas *Aug 2024 – Present*

- Lead a 20+ member executive team and manage a $1,000+ annual budget, growing participation 30% across 5+ cultural events.

Team Leader, The Big Event – University of Central Arkansas *Mar 2024 – Present*

- Directed 20+ volunteers across community service projects totaling 50+ hours of service.